something related, yet richer: an object's combined motion through space and time. The key fact, Einstein discovered, is that these two kinds of motion are always complementary. When the parked car you were looking at speeds away, what really happens is that some of its light-speed motion is diverted from motion through time into motion through space, *keeping their combined total unchanged.* Such diversion unassailably means that the car's motion through time slows down.

As an example, if Lisa had been able to see Bart's watch as he sped along at 500 million miles per hour, she would have seen that it was ticking about two-thirds as fast as her own. For every three hours that passed on Lisa's watch, she would see that only two had passed on Bart's. His rapid motion through space would have proved a significant drain on his speed through time.

Moreover, the maximum speed through space is reached when all light-speed motion through time is fully diverted into light-speed motion through space—one way of understanding why it is impossible to go through space at greater than light speed. Light, which always travels at light speed through space, is special in that it always achieves such total diversion. And just as driving due east leaves no motion for traveling north, moving at light speed through space leaves no motion for traveling through time! Time stops when traveling at the speed of light through space. A watch worn by a particle of light would not tick at all. Light realizes the dreams of Ponce de León and the cosmetics industry: it doesn't age.[5]

As this description makes clear, the effects of special relativity are most pronounced when speeds (through space) are a significant fraction of light speed. But the unfamiliar, complementary nature of motion through space and time always applies. The lesser the speed, the smaller the deviation from prerelativity physics—from common sense, that is—but the deviation is still there, to be sure.

Truly. This is not dexterous wordplay, sleight of hand, or psychological illusion. This is how the universe works.

In 1971, Joseph Hafele and Richard Keating flew state-of-the-art cesium-beam atomic clocks around the world on a commercial Pan Am